

The Co-Pilot That Says No

A mission guard that allows normal flight and vetoes the commands that cross a hard line.

SIM-ONLY

— THE PROBLEM

An autonomy stack flies what it's told

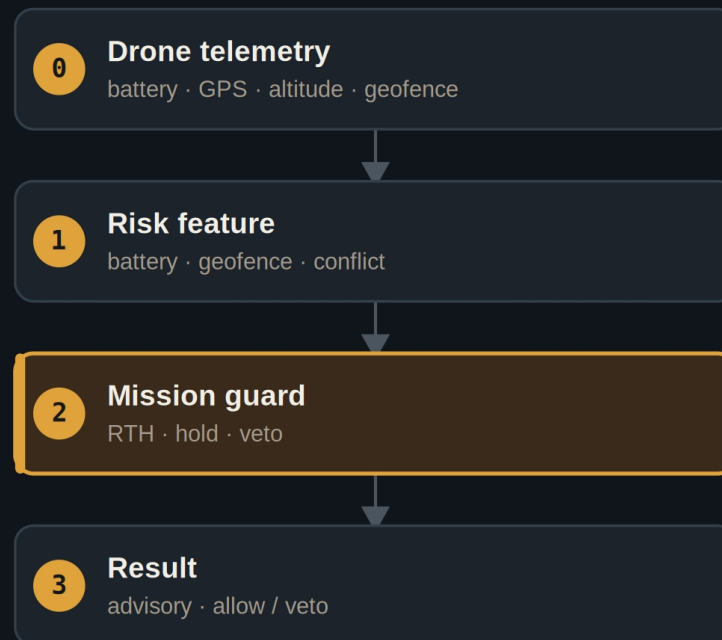
- The mission planner will happily route through a no-fly boundary if the path is shorter.
- It will keep flying on a dying battery until it can't get home.
- When that ends badly, there's no record of what should have stopped it.

The aircraft doesn't need more autonomy. It needs a veto.

A guard that vets every command

- **It watches**
battery, GPS quality, altitude, and geofence distance, every tick.
- **It allows**
ordinary commands pass straight through — no drag on the mission.
- **It refuses**
a geofence-breaching command is vetoed; a low battery calls return-to-home.

Telemetry in, allow-or-veto out



The mission-guard layer holds the veto and the return-to-home recommendation.

Permissive by default, decisive at the line

ALLOWED

- 22 commands passed through.
- Normal flight is never slowed.

VETOED

- 2 geofence-breaching commands stopped at t6.
- Low battery at t12 → return-to-home.

50 deterministic ticks

22 : 2

commands allowed vs vetoed
permissive, then firm

t6

geofence breach vetoed before effect

t12

low battery → return-to-home

Sim-only until it's earned otherwise

SIM-ONLY Vetoes and recommendations affect only the simulation — no wire to a flight controller.

Permissive Allows ordinary flight; intervenes only at a hard boundary.

Auditable Every allow and veto is on the record.

Toward a real platform

- **Swap the input edge**
the mock telemetry becomes a MAVLink listener.
- **Keep the contract**
same typed signals, same veto logic.
- **Certify deliberately**
nothing leaves the simulation until the integration is signed off.

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The safest autonomous aircraft isn't the one that flies best — it's the one with a veto you can read after the fact.

The case for Demo 03

JNEOPALLIUM · DEMO 03

rakovpublic/jneopallium

A co-pilot that knows the line.

Permissive on normal flight, decisive at a geofence — on the record.

github.com/rakovpublic/jneopallium

SIM-ONLY · deterministic

